

Daily Combinatorics Drill #4 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Binomial Theorem · Pascal's Rule · Coefficient Extraction · Substituting Values into Expansions.

Section A Rapid Recognition

(≤15 s each)

1. Coefficient of x^2 in $(1+x)^5$.

Answer: $\binom{5}{2} = 10$

Method: Binomial theorem: the x^k coefficient in $(1+x)^n$ is $\binom{n}{k}$.

Investigate further: Why is it $\binom{5}{2}$? Expanding $(1+x)(1+x)(1+x)(1+x)(1+x)$, an x^2 term arises by choosing the x from exactly 2 of the 5 brackets. The binomial theorem is a counting statement — that lens powers everything on this sheet.

2. Coefficient of x in $(x+2)^3$.

Answer: 12

Method: Term with x^1 : $\binom{3}{1}x \cdot 2^2 = 12x$. Keep track of which quantity is being raised to which power.

Investigate further: Write out all four terms of $(x+2)^3$ and check they sum to 27 at $x=1$. The “evaluate at $x=1$ ” check catches most expansion slips in exams.

3. Row 5 of Pascal's triangle.

Answer: 1, 5, 10, 10, 5, 1

Method: Adjacent sums of row 4 (1, 4, 6, 4, 1), or $\binom{5}{k}$ directly.

Investigate further: Memorise rows 0–8 cold. Then look at the triangle mod 2 (odd entries only) — shade them for rows 0–16 and meet the Sierpiński triangle. D3 explains the pattern.

4. Coefficient of x^3y^2 in $(x+y)^5$.

Answer: $\binom{5}{2} = 10$

Method: Choose which 2 of the 5 brackets contribute y : $\binom{5}{2}$.

Investigate further: Coefficients of x^3y^2 and x^2y^3 are equal — symmetry of $\binom{5}{2} = \binom{5}{3}$ made visible. In how many terms does $(x+y)^5$ expand? (6 — one per power split.)

5. Sum of all coefficients of $(1+x)^7$.

Answer: $2^7 = 128$

Method: Substitute $x = 1$: the expansion collapses to its coefficient sum.

Investigate further: Substituting values into an identity is the sheet's master move. What does $x = 2$ give? (3^7 — the sum $\sum \binom{7}{k} 2^k$.) Every substitution is a new identity for free.

6. Constant term of $(x + \frac{1}{x})^4$.

Answer: $\binom{4}{2} = 6$

Method: Powers cancel when the x 's and $\frac{1}{x}$'s balance: 2 of each.

Investigate further: Why does $(x + \frac{1}{x})^5$ have *no* constant term? (Odd total degree — balance is impossible.) Parity kills entire terms before any computation.

7. Express $\binom{9}{4} + \binom{9}{5}$ as one coefficient.

Answer: $\binom{10}{5}$ (which is 252)

Method: Pascal's rule: adjacent entries sum to the entry below. No evaluation needed — the recognition is the answer.

Investigate further: Prove it in one line with committees (a committee of 5 from 10 either contains the distinguished person or not) — you will need exactly this argument in C8.

8. $\binom{6}{0} - \binom{6}{1} + \binom{6}{2} - \cdots + \binom{6}{6}$.

Answer: 0

Method: Substitute $x = -1$ into $(1+x)^6$: $0^6 = 0$.

Investigate further: Consequence: in any row, the even-index entries and odd-index entries have equal sums (each 2^{n-1} — B5 uses this). Pair with sheet 2's subset-parity bijection: same fact, two proofs.

9. True or false: coefficient of x^2 in $(1-x)^{10}$ is negative.

Answer: False — it is +45

Method: The x^2 term is $\binom{10}{2}(-x)^2 = 45x^2$: the square eats the sign. Signs alternate, and even powers are positive.

Investigate further: TMUA-style sign traps live exactly here. Which coefficients of $(1-x)^{10}$ are negative? (The odd ones.) State the general sign rule for $(a-b)^n$ and never lose a mark to it again.

10. Largest coefficient in $(1+x)^4$.

Answer: $\binom{4}{2} = 6$

Method: Rows of Pascal's triangle rise to the middle and fall symmetrically.

Investigate further: For even n there is one maximum; for odd n , two equal ones. Prove the rise-then-fall by examining the ratio $\binom{n}{k+1}/\binom{n}{k} = \frac{n-k}{k+1}$ — when does it drop below 1? (C2 uses this.)

Section B Manipulation Drills

(≤50 s each)

1. Coefficient of x^3 in $(2 + 3x)^5$.

Answer: $\binom{5}{3} 2^2 3^3 = 10 \cdot 4 \cdot 27 = 1\,080$

Method: General term $\binom{5}{k} 2^{5-k} (3x)^k$; set $k = 3$. Both constants carry powers — forgetting 2^{5-k} is the classic slip.

Investigate further: Check digit-cheaply: at $x = 1$ the whole expansion is $5^5 = 3125$; your five coefficients must sum to it. Run the check — it takes 20 seconds and catches the missing-power error instantly.

2. Coefficient of x^5 in $(1 + 2x)^4(1 + x)^3$.

Answer: 168

Method: Convolve the tails only: from $(1 + 2x)^4$ take coefficients 16, 32, 24 (of x^4, x^3, x^2) and from $(1 + x)^3$ take 3, 3, 1 (of x, x^2, x^3): $16 \cdot 3 + 32 \cdot 3 + 24 \cdot 1 = 168$.

Investigate further: Had both brackets shared the base $(1 + x)$, you would merge first: $(1 + x)^4(1 + x)^3 = (1 + x)^7$, coefficient $\binom{7}{5} = 21$ in five seconds. Check the bases before you convolve — and note that separate-then-convolve on the shared-base version proves $\sum_k \binom{4}{k} \binom{3}{5-k} = \binom{7}{5}$, a Vandermonde identity (D2).

3. Term independent of x in $(2x + \frac{1}{x^2})^6$.

Answer: $\binom{6}{2} 2^4 = 240$

Method: General term: $\binom{6}{k} (2x)^{6-k} x^{-2k}$ has degree $6 - 3k$. Zero degree forces $k = 2$: $\binom{6}{2} 2^4$.

Investigate further: Degree bookkeeping first, arithmetic second. Which powers of x *can* occur here? ($6 - 3k$: only 6, 3, 0, -3, -6, -9, -12 — gaps of 3.) Knowing the degree lattice tells you instantly that e.g. an x^2 term is impossible.

4. Coefficient of x^2 in $(1 + x)^n$ is 66. Find n .

Answer: $n = 12$

Method: $\binom{n}{2} = 66 \Rightarrow n(n-1) = 132 = 12 \cdot 11$.

Investigate further: Sheet 2's consecutive-product reflex again. Harder inverse: the coefficient of x^3 is 220 — find n ($n(n-1)(n-2) = 1320 = 12 \cdot 11 \cdot 10$). Estimation ($\sqrt[3]{1320} \approx 11$) then check.

5. Coefficient of x^6 in $(x^2 + \frac{1}{x})^6$.

Answer: $\binom{6}{2} = 15$

Method: General term $\binom{6}{k}(x^2)^{6-k}x^{-k}$ has degree $12 - 3k$; degree 6 forces $k = 2$.

Investigate further: The degree lattice here is 12, 9, 6, 3, 0, -3, -6. Which is the constant term? ($k = 4$: $\binom{6}{4} = 15$ — equal to the x^6 coefficient. Coincidence, or a symmetry of this particular bracket? Substitute $x \rightarrow \frac{1}{x}$ and find out.)

6. Coefficient of x^2 in $(1 + 2x)^4(1 - x)^2$.

Answer: 9

Method: Convolves only what is needed: x^2 coefficient = $24 \cdot 1 + 8 \cdot (-2) + 1 \cdot 1 = 9$, using $(1 + 2x)^4 = 1 + 8x + 24x^2 + \dots$ and $(1 - x)^2 = 1 - 2x + x^2$.

Investigate further: Never expand whole products for one coefficient — harvest each factor only up to the degree you need. Practise: coefficient of x^3 in the same product ($32 \cdot 1 + 24 \cdot (-2) + 8 \cdot 1 = -8$).

7. Evaluate 11^3 and 9^3 by binomial.

Answer: $11^3 = 1331$; $9^3 = 729$

Method: $(10 + 1)^3 = 1000 + 300 + 30 + 1$; $(10 - 1)^3 = 1000 - 300 + 30 - 1$. Pascal row 3 does the arithmetic.

Investigate further: Notice 11^3 's digits ARE row 3 of Pascal's triangle (1,3,3,1) — and $11^4 = 14641$ is row 4. When does the pattern break, and why? (Row 5 has a two-digit entry: carrying destroys the display.)

8. Coefficient of x^3 in $(1 + kx)^6$ is 160. Find k .

Answer: $k = 2$

Method: $\binom{6}{3}k^3 = 20k^3 = 160 \Rightarrow k^3 = 8$.

Investigate further: Could a negative or fractional k ever be the intended answer in such questions? Solve $\binom{6}{3}k^3 = -540$ and $\binom{4}{2}k^2 = \frac{3}{2}$ to see both arise naturally in TMUA-style settings.

9. $\binom{n}{2} = \binom{n}{3}$. Find n .

Answer: $n = 5$

Method: Equal entries sit symmetrically: $\binom{n}{r} = \binom{n}{s}$ (with $r \neq s$) forces $r + s = n$. So $n = 2 + 3 = 5$.

Investigate further: Solve without symmetry, by ratio: $\frac{\binom{n}{3}}{\binom{n}{2}} = \frac{n-2}{3} = 1$. The ratio form generalises to “for which k is $\binom{n}{k}$ maximal?” — close the loop with A10.

10. Coefficient of a^2b^2c in $(a + b + c)^5$. (cf. TMUA 2020 P1 Q13)

Answer: $\frac{5!}{2!2!1!} = 30$

Method: Multinomial: distribute the 5 brackets among the letters — exactly the BANANA formula from sheet 3 wearing algebra.

Investigate further: Repeated-letter words, flag rows, multinomial coefficients: one formula, three sheets. How many *distinct monomials* does $(a + b + c)^5$ have? (Solutions of $i + j + k = 5$: that is sheet 5's stars-and-bars, arriving tomorrow.)

Section C Substitution & Structure

(≤90 s each)

1. $\binom{10}{0} + \binom{10}{2} + \cdots + \binom{10}{10}.$

Answer: $2^9 = 512$

Method: Substitute twice into $(1 + x)^{10}$: $x = 1$ gives 2^{10} (all terms), $x = -1$ gives 0 (alternating). Adding kills the odd-index terms and doubles the even ones: $\text{sum} = (2^{10} + 0)/2 = 512$.

Investigate further: The add/subtract-substitutions trick extracts any alternating pattern. Try extracting $\binom{10}{0} + \binom{10}{4} + \binom{10}{8}$ — what would you substitute? (Fourth roots of unity: a genuine BMO-level extension to sit with.)

2. Greatest coefficient in $(1 + x)^8$.

Answer: $\binom{8}{4} = 70$

Method: Ratio test from A10's inv: $\frac{\binom{8}{k+1}}{\binom{8}{k}} = \frac{8-k}{k+1} > 1$ exactly while $k < 3.5$, so coefficients climb to $k = 4$ then fall.

Investigate further: Now find the greatest coefficient of $(1 + 2x)^8$ — the ratio becomes $\frac{2(8-k)}{k+1}$ and the peak moves right (to $k = 5$: check it). Peaks shift towards the larger term; TMUA loves this.

3. Estimate 1.01^{10} from three terms.

Answer: $1 + 10(0.01) + 45(0.01)^2 = 1.1045$

Method: $(1 + x)^{10}$ at $x = 0.01$; terms shrink so fast the tail is invisible at 4 d.p.

Investigate further: Bound the error: the next term is $120(0.01)^3 = 1.2 \times 10^{-4}$... which *would* touch the 4th decimal place. Investigate: the true value is 1.10462..., so why did the estimate still round correctly? (The tail is positive but small enough. Always *bound*, not assume.)

4. In $(1 + 3x)^n$, coefficient of x^5 is three times that of x^4 . Find n .

Answer: $n = 9$

Method: $\binom{n}{5}3^5 = 3 \cdot \binom{n}{4}3^4$. The powers of 3 cancel completely (3^5 on both sides), leaving $\binom{n}{5} = \binom{n}{4}$, so $n = 4 + 5 = 9$ by symmetry.

Investigate further: The constants were a decoy — strip them before solving. Equal *adjacent* plain coefficients happen only for odd n at the centre (A10's two equal maxima). For a genuine constant

effect, solve: in $(1 + 2x)^n$ the x^5 coefficient is *four* times the x^4 one; now the 2's don't cancel.

5. Coefficient of x^4 in $(1 + x)^5(1 - 2x)^3$. (cf. MAT 2023 Q1F)

Answer: 25

Method: Convolution with signs, harvesting only degree-4 pairs: $5 \cdot 1 + 10 \cdot (-6) + 10 \cdot 12 + 5 \cdot (-8) = 5 - 60 + 120 - 40 = 25$. Lay the two coefficient lists side by side and zip them — do not expand the product.

Investigate further: MAT's Q1 multiple-choice regularly hides one sign or one binomial power slip among its options. Recompute with the factors swapped in your zip to double-check — convolution is symmetric, so any asymmetry in your two answers locates the error.

6. $\sum_{k=0}^5 k \binom{5}{k}$.

Answer: $5 \cdot 2^4 = 80$

Method: Committee-chair (sheet 2, B5): $k \binom{5}{k} = 5 \binom{4}{k-1}$, and summing the right side gives $5 \cdot 2^4$. Alternatively differentiate $(1 + x)^5$ and set $x = 1$.

Investigate further: Two proofs from two different sheets — the counting one and the calculus one. Use either to evaluate $\sum k^2 \binom{5}{k}$ (answer $5 \cdot 6 \cdot 2^3 =$ check it: $k^2 = k(k-1) + k$). Identities stack.

7. Evaluate $\sum_{k=0}^6 \binom{6}{k} 2^k$ by substitution.

Answer: $3^6 = 729$

Method: The sum is exactly $(1 + x)^6$ at $x = 2$: $(1 + 2)^6 = 729$. Recognising a binomial expansion *in reverse* is the skill — the 2^k pattern names the substitution.

Investigate further: Every value of x turns $(1 + x)^n$ into an identity: $x = 10$ explains why row- n digits “read as” 11^n (B7), and $x = -\frac{1}{2}$ gives an alternating shrinking sum. Evaluate $\sum \binom{6}{k} (-1)^k 2^{6-k}$ by spotting *its* disguise $((2 - 1)^6 = 1)$.

8. Prove Pascal's rule by committee; build row 7 from row 6.

Answer: Row 7: 1, 7, 21, 35, 35, 21, 7, 1

Method: *Proof.* A committee of r from n people either includes a distinguished person P or not. Including P : $\binom{n-1}{r-1}$ choices for the rest. Excluding P : $\binom{n-1}{r}$. These cases are disjoint and exhaustive, so $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$. $\square$ Adjacent sums of row 6 (1, 6, 15, 20, 15, 6, 1) then give row 7.

Investigate further: This one-sentence proof is a model of BMO1 writing: cases named, disjointness noted, conclusion drawn. Rewrite sheet 2 B9's Priya split as a formal proof in the same register — Section D now expects this standard.

Section D Challenge

(SMC / BMO1 difficulty)

1. Prove $\binom{2n}{n}$ is even for all $n \geq 1$.

Answer: Proof below.

Method: *Proof.* By Pascal's rule, $\binom{2n}{n} = \binom{2n-1}{n-1} + \binom{2n-1}{n}$. By the symmetry $\binom{m}{r} = \binom{m}{m-r}$ applied with $m = 2n - 1$: $\binom{2n-1}{n-1} = \binom{2n-1}{(2n-1)-(n-1)} = \binom{2n-1}{n}$. So $\binom{2n}{n}$ is the sum of two *equal* integers, hence even. $\square$

Method: *Alternative (bijective) proof.* Pair each n -subset S of $\{1, \dots, 2n\}$ with its complement S^c . Since $|S| = n = |S^c|$ and $S \neq S^c$ always, the n -subsets split into disjoint pairs, so their number is even. $\square$

Investigate further: Two complete written proofs — study how each names its key step. Sharper question: what is the largest power of 2 dividing $\binom{2n}{n}$? Compute it for $n = 1, \dots, 8$ and conjecture (Kummer's theorem: the number of carries when adding $n + n$ in binary).

2. Show $\sum_{k=0}^{10} \binom{10}{k}^2 = \binom{20}{10}$.

Answer: Both count $\binom{20}{10} = 184\,756$.

Method: *Proof.* Consider choosing 10 people from a room of 10 women and 10 men: $\binom{20}{10}$ ways. Alternatively, classify each choice by the number k of women chosen: $\binom{10}{k}$ ways for the women and $\binom{10}{10-k}$ for the men, and $\binom{10}{10-k} = \binom{10}{k}$ by symmetry. Summing the disjoint classes: $\sum_k \binom{10}{k}^2$. Two counts of one set are equal. $\square$

Investigate further: This is Vandermonde's identity $\sum_k \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$ at $m = n = r = 10$ — sheet 2's C5 inv, now proved properly. Also prove it by extracting the x^{10} coefficient from $(1+x)^{10}(1+x)^{10}$: the algebra proof and the committee proof should feel like translations of each other.

3. Prove exactly four of the $\binom{12}{k}$ are odd. (cf. TMUA 2023 P1 Q6)

Answer: The odd ones are exactly $k = 0, 4, 8, 12$.

Method: *Proof.* All congruences are mod 2. First, $(1+x)^2 = 1 + 2x + x^2 \equiv 1 + x^2$, and squaring again, $(1+x)^4 \equiv (1+x^2)^2 \equiv 1 + x^4$. Hence $(1+x)^{12} = ((1+x)^4)^3 \equiv (1+x^4)^3 = 1 + 3x^4 + 3x^8 + x^{12} \equiv 1 + x^4 + x^8 + x^{12}$. Comparing coefficients of x^k on both sides: $\binom{12}{k}$ is odd precisely when $k \in \{0, 4, 8, 12\}$ — four values. $\square$

Investigate further: General law (Lucas): $\binom{n}{k}$ is odd iff every binary digit of k is $\leq$ the corresponding digit of n . Here $n = 12 = 1100_2$, so $k \in \{0000, 0100, 1000, 1100\}$ — the same four. Verify Lucas on row 10 (1010_2 : which k ?), and connect to A3's Sierpiński picture.

4. Show $\binom{2}{2} + \binom{3}{2} + \dots + \binom{9}{2} = \binom{10}{3}$.

Answer: $\binom{10}{3} = 120$

Method: *Proof (classify by largest element).* Count the 3-subsets of $\{1, 2, \dots, 10\}$: there are $\binom{10}{3}$. Classify each subset by its largest element $m+1$ (so $m+1$ ranges over $3, \dots, 10$): the other two elements are then any 2-subset of $\{1, \dots, m\}$, giving $\binom{m}{2}$ subsets in that class. The classes are disjoint and exhaustive, so $\sum_{m=2}^9 \binom{m}{2} = \binom{10}{3}$. $\square$

Investigate further: This is the *hockey stick identity*: any diagonal run in Pascal's triangle sums to the entry below-right of its end. Prove it a second way by iterating Pascal's rule. Then use it to re-derive $1 + 2 + \cdots + n = \binom{n+1}{2}$ and $\sum m^2$ (write $m^2 = 2\binom{m}{2} + \binom{m}{1}$).

5. p prime, $0 < k < p$: prove $p \mid \binom{p}{k}$.

Answer: Proof below.

Method: *Proof.* Write $k! \binom{p}{k} = p(p-1) \cdots (p-k+1)$. The right side is visibly divisible by p , so $p \mid k! \binom{p}{k}$. Since p is prime and every factor of $k! = 1 \cdot 2 \cdots k$ is less than p , we have $p \nmid k!$, so p and $k!$ are coprime. A prime dividing a product must divide one of the factors; as it cannot divide $k!$, it divides $\binom{p}{k}$. $\square$ (Note where primality is used — *twice*.)

Investigate further: Consequence: $(1+x)^p \equiv 1+x^p \pmod{p}$ — the “freshman's dream” is a theorem mod p . Feed $x=1$ into it to get $2^p \equiv 2 \pmod{p}$: you have proved Fermat's little theorem for base 2. Test the failure for composites: is $\binom{4}{2}$ divisible by 4? This lemma returns as a workhorse in the number-theory pillar.

Top 5 Patterns Today

- The general term.** Write $\binom{n}{k} a^{n-k} b^k$ with the constants inside, solve the degree equation for k , then compute. Never expand what you can index.
- Substitute values into identities.** $x=1$: coefficient sum. $x=-1$: alternating sum. Add/subtract the two: even/odd extraction. Every substitution is a theorem.
- Pascal's rule = in-or-out.** $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$ is the committee split from sheet 2, and it generates the whole triangle.
- Count one thing two ways.** Vandermonde, hockey stick, committee-chair: pick the right scenario and the identity proves itself in a sentence.
- Binomials mod small primes.** $(1+x)^2 \equiv 1+x^2 \pmod{2}$ propagates to $(1+x)^{2^m} \equiv 1+x^{2^m}$: parity patterns of whole rows fall out (Lucas/Sierpiński).

Common Traps to Avoid

- Dropping the constant's power.** In $(2+3x)^5$ the 2^{5-k} is compulsory. Coefficient-sum check at $x=1$ catches this in seconds.
- Sign slips with $(a-b)^n$.** Keep $(-b)^k$ intact inside the general term; decide the sign from the parity of k , never by feel.
- Expanding both factors fully for one coefficient.** Harvest each factor only to the degree needed and zip. Full expansion is slow and error-prone.
- Assuming symmetric-looking sums need algebra.** $\sum \binom{10}{k}^2$ falls to a story about choosing 10 people. Reach for the committee before the formula sheet.
- Using primality nowhere.** If your proof of D5 never invokes that p is prime, it proves the false statement $4 \mid \binom{4}{2}$. Locate where every hypothesis earns its keep.

“An identity is a story told twice — once for each side of the equals sign.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths